

BOGORODSKOYE, Russian Federation

WMO: 314390

Lat: **52.3731N**Lon: **140.4330E**Elev: **35**StdP: **100.91**Time Zone: **10.00 (E10)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.4	-32.4	-37.2	0.1	-34.2	-35.1	0.1	-32.3	12.4	-15.0	10.4	-15.8	0.6	160	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	27.0	20.4	25.5	19.4	24.0	18.5	21.7	25.6	20.5	24.1	19.4	23.0	2.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	15.0	24.4	19.0	13.9	22.8	17.9	12.9	21.7	63.7	25.5	59.4	24.3	55.6	23.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.4	6.2	-37.1	30.2	2.7	1.8	-39.1	31.5	-40.7	32.6	-42.3	33.6	-44.3	35.0		
(5)			-36.7	23.6	3.1	1.6	-38.9	24.8	-40.7	25.7	-42.5	26.7	-44.7	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-22.4	-19.3	-10.4	0.0	8.3	14.7	18.4	17.8	12.0	3.1	-9.9	-20.3		
	DBStd	15.45	6.30	5.59	5.24	4.23	3.94	3.64	3.00	2.67	3.60	3.96	6.78	6.65		
	HDD10.0	4620	1006	821	632	302	81	5	0	0	21	216	597	941		
	HDD18.3	6977	1264	1054	891	551	311	118	36	43	191	472	847	1199		
	CDD10.0	760	0	0	0	0	28	145	262	242	81	2	0	0		
	CDD18.3	77	0	0	0	0	0	9	40	27	1	0	0	0		
	CDH23.3	471	0	0	0	0	7	87	245	128	4	0	0	0		
	CDH26.7	61	0	0	0	0	1	10	40	11	0	0	0	0		
(14) Wind	WSAvg	2.3	2.3	2.5	2.6	2.5	2.6	2.1	2.0	1.9	1.9	2.3	2.3	2.2		
(15) Precipitation	PrecAvg	520	27	16	19	25	47	52	56	96	61	52	35	35		
	PrecMax	705	64	54	43	49	88	161	120	190	132	128	81	92		
	PrecMin	277	1	1	2	6	11	9	1	6	3	9	8	3		
	PrecStd	98	17	13	13	11	24	34	32	48	28	29	18	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	-2.5	4.2	15.4	24.0	27.9	29.7	28.0	23.9	16.1	5.5	-0.3		
		MCWB	-1.8	-3.6	0.5	8.2	14.4	19.6	22.1	22.3	18.2	10.9	3.6	-1.1		
	2%	DB	-4.8	-5.8	1.4	11.2	20.7	25.6	27.7	26.2	21.8	13.0	2.8	-2.8		
		MCWB	-5.4	-6.8	-1.2	5.6	12.7	18.2	20.7	20.8	17.2	9.5	1.2	-3.4		
	5%	DB	-9.5	-8.5	-0.3	8.4	18.3	23.8	25.9	24.8	19.8	10.8	1.1	-6.2		
		MCWB	-10.0	-9.4	-2.3	3.9	11.1	17.1	19.6	19.9	15.7	7.7	-0.3	-6.6		
	10%	DB	-12.9	-10.8	-1.9	6.2	15.9	22.0	24.4	23.2	17.9	9.0	-0.4	-10.8		
		MCWB	-13.5	-11.6	-3.6	2.5	9.8	16.0	18.9	18.9	14.7	6.5	-1.7	-11.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.9	-3.6	0.9	8.7	15.5	21.2	23.5	23.4	19.4	11.9	3.7	-0.9		
		MCDB	-1.3	-2.8	3.4	14.1	22.1	26.4	28.2	26.6	22.4	14.5	5.2	-0.3		
	2%	WB	-5.5	-6.9	-0.8	6.1	13.4	19.2	21.8	21.9	17.6	9.9	1.5	-3.5		
		MCDB	-5.0	-5.9	1.1	10.5	19.5	24.0	26.3	25.1	20.9	12.4	2.7	-2.8		
	5%	WB	-10.0	-9.3	-2.2	4.2	11.8	17.9	20.6	20.6	16.4	8.2	-0.1	-6.6		
		MCDB	-9.5	-8.5	-0.5	7.9	16.7	22.6	24.7	23.6	19.2	10.3	1.0	-6.2		
	10%	WB	-13.4	-11.7	-3.6	2.7	10.4	16.7	19.5	19.5	15.0	6.7	-1.7	-11.3		
		MCDB	-12.9	-10.8	-1.9	5.8	15.1	21.1	23.4	22.5	17.6	8.7	-0.6	-10.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.5	10.4	9.0	10.1	10.2	8.8	8.3	8.9	7.6	7.3	7.7		
		MCDBR	7.8	9.4	10.6	12.5	15.5	13.9	12.1	11.2	11.6	10.7	6.8	8.2		
		MCWBR	7.4	8.6	8.4	7.7	8.4	7.6	6.5	6.4	7.1	7.0	5.5	8.1		
		MCDBR	7.4	9.0	9.8	12.0	13.6	12.4	10.6	9.9	10.4	9.9	6.2	8.3		
	5% WB	MCWBR	7.3	8.3	7.9	7.6	8.0	7.3	6.4	6.1	6.9	7.0	5.4	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.280	0.310	0.381	0.489	0.551	0.521	0.520	0.448	0.385	0.390	0.336	0.288		
	taud		2.052	2.004	1.849	1.715	1.768	1.942	1.995	2.224	2.339	2.125	2.048	2.013		
	Ebn at Noon		680	772	776	734	713	744	737	777	789	672	588	581		
	Edh at Noon		77	111	160	209	208	177	165	125	99	99	79	68		
(44) All-Sky Solar Radiation	RadAvg		1.22	2.39	3.79	4.63	4.84	5.38	4.82	4.07	3.26	1.98	1.17	0.89		
	RadStd		0.10	0.13	0.26	0.39	0.52	0.58	0.60	0.39	0.28	0.17	0.14	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-138	+61	N/A

Nomenclature: See separate page